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Cholera surveillance and estimation of burden of cholera

Divya Ganesan [a,1], Sanjukta Sen Gupta [b], Dominique Legros [c,*]

[a] Global Health Strategies, 18/1, Qutab Institutional Area, New Delhi 110016, India
[b] Translational Health Science and Technology Institute, NCR Biotech Science Cluster, 3rd Milestone, Faridabad – Gurgaon Expressway, Faridabad, Haryana 121001, India
[c] The Global Task Force on Cholera Control, World Health Organization, Avenue Appia 20, CH-1211 Geneva 27, Switzerland

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ABSTRACT

Cholera continues to be poorly controlled in multiple epidemic and endemic areas across the globe, with estimated annual incidence of 1.3–4.0 million cases, resulting in 21,000 to 143,000 deaths worldwide in 2015. The usual approach for patient diagnosis and cholera surveillance is clinical examination of cases of acute watery diarrhea (AWD), confirmed by positive culture or polymerase chain reaction tests. Rapid diagnostic tests (RDTs) are used in regions with limited laboratory capacities but have been found to demonstrate large variations in performance, ranging in sensitivity from 58% to 100% and in specificity from 60% to 100%. Most countries rely on hospital-based surveillance of diarrheal disease to compute the cholera burden. The World Health Organization (WHO) recommends that countries assess public health events involving cholera against the International Health Regulations 2005 criteria and determine need for official notification using the standard case definition. Cholera is an often under-recognized and under reported problem because of differences in case definitions, reluctance by authorities to acknowledge and report cholera, inadequacies in hospital surveillance systems, lack of effective diagnostic tests and commonalities in clinical presentation of cholera with other AWD etiologies. The resulting gap in burden data impairs economic analysis of disease impact and identification of areas for targeted control interventions. There is an urgent need to strengthen surveillance data by supplementing reported numbers with estimates from literature reviews and data from modelling studies, developing better-performing RDTs, enhancing monitoring and evaluation processes of in-country surveillance systems, and encouraging countries to report cholera cases by "rewarding" better reporting with technical support and improved access to vaccines. It is imperative that immediate steps are taken towards strengthening surveillance and reporting systems globally, especially in cholera-prone and resource-limited areas, where it will enable countries to articulate their demand for resources more accurately.
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1. Introduction

Cholera is a bacterial infection of humans caused by *V. cholerae*, a Gram-negative bacillus divided into more than 200 different serogroups [1]. *V. cholerae* serogroup O1, biotype El Tor, serotype Ogawa or Inaba, is the principal pathogen. In 1993, a new serogroup *V. cholerae* O139 was reported from southern India and has been responsible for outbreaks in Bangladesh and Thailand. Infection occurs following ingestion of food or water contaminated by faeces containing high concentrations of *V. cholerae*. Transmission may occur through two routes: a faecal–oral route ('person to faeces to person') or through direct infection from the environment (environmentally acquired infection). Only a minority (10–20%) of cholera infections are symptomatic and present as mild acute watery diarrhoea (AWD) that can be treated with oral rehydration solutions. About 20% of symptomatic cases are severe, characterized by the abrupt onset of profuse watery diarrhoea and vomiting resulting in dehydration. Without rapid and appropriate intravenous rehydration, correction of electrolyte imbalance and antibiotics, severe cholera cases can result in death very quickly [2].

Cholera remains badly controlled in multiple epidemic and endemic areas across the globe. Cholera occurs endemically in south and Southeast Asia and in sub-Saharan Africa and has recently returned to the Americas with the ongoing transmission in Haiti [2]. In 2015, 42 countries reported a total of 172,454 cases of cholera, including 1304 deaths, to WHO [3]. However, the estimated annual incidence of cholera is 1.3–4.0 million cases,

* Corresponding author at: World Health Organization, Avenue Appia 20, CH-1211 Geneva 27, Switzerland.
E-mail address: legrosd@who.int (D. Legros).
[1] Present address: Emory University Rollins School of Public Health, 1518 Clifton Rd, Atlanta, GA 30322, USA.

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resulting in 21,000–143,000 deaths worldwide. About 1.3 billion people are still potentially at risk for cholera in endemic countries [4].

2. Burden estimation and surveillance

2.1. Current surveillance and reporting structure

Disease surveillance is essential for determining burden of disease in any setting, for measuring changes in morbidity and mortality patterns, detecting outbreaks early in time and designing, implementing and monitoring suitable responses to them. The usual approach for patient diagnosis and for surveillance of cholera is based on clinical examination of cases of AWD, suspected of cholera. Positive culture or polymerase chain reaction (PCR) tests performed on stool samples in reference laboratories are needed for confirmation, due to overlap in clinical presentation of cholera with other enteric diseases. In regions with limited laboratory capacities, rapid diagnostic tests (RDTs) are used to detect suspected cholera cases among all AWD. However, evaluations of the available RDTs have found large variations in performance ranging in sensitivity from 58% to 100% and in specificity from 60% to 100% [5]. While these tests are designed to be used in field conditions to enable rapid implementation of control measures, stool culture or PCR of suspected samples remains the "gold standard" for cholera diagnosis and for confirmation of positive RDTs. The use of filter paper is a low-cost and low-maintenance alternative to the traditional Cary-Blair transport media for preserving and transporting samples to laboratories with the capability to conduct PCR and will help enhance its reach and utility [6]. Molecular surveillance through use of isolated *V. cholerae* strains or PCR amplicons can be enhanced by addition of strong bioinformatics capacity.

Most countries rely on hospital-based surveillance of diarrhoeal disease to compute the cholera burden. Countries in the WHO-Africa region adopted the Integrated Disease Surveillance and Response (IDSR) strategy in 1998 to enhance surveillance for priority public health diseases, conditions and events, including cholera. Each national IDSR strategy is based on its own disease priorities and administrative capacities and processes [7]. For example, in Ghana, the IDSR reporting is through the internet-based District Health Information Management System 2 (DHIMS2) and follows a hierarchical order starting at peripheral health facilities that report to district health centres, followed by regional and national levels [8]. In Uganda, IDSR includes both private and public sector. The levels of the health information system are community/health facilities, district and national, and each has fixed weekly and monthly reporting schedules [9].

India established the Integrated Disease Surveillance Programme (IDSP) in 2004, which covers surveillance and reporting of a few infectious diseases including cholera. The administrative mechanism of the IDSP involves surveillance units at the district, state and national levels and integrates the private and public-sector hospitals at the state level. Data collection, compilation, analysis and reporting are by computerized systems down to the primary health-care level [10]. Philippines Department of Health adopted the Philippine Integrated Disease Surveillance and Response (PIDSR) system in 2007 to strengthen epidemiological surveillance and response in compliance with International Health Regulations (IHR) 2005 requirements. In this system, Disease Reporting Units (DRU) include Barangay Health Stations (BHS), Rural Health Units (RHU) and City Health Offices (CHU) and government and private hospitals, clinics and laboratories. These detect and record notifiable diseases in one of three PIDSR forms which are passed to the next levels, namely, Provincial/City Epidemiology and Surveillance Units (PESU/CESU), Regional Epidemiology and Surveillance Unit (RESU) and the National Epidemiology Centre, electronically or in paper form [11]. Cholera is a Category II weekly notifiable disease in the PIDSR [12].

2.2. Who's global data sourcing/estimation systems

Notification of cases of cholera to WHO is no longer mandatory, under the IHR of 2005. However, WHO recommends that countries assess public health events involving cholera against the criteria provided in the regulations and determine whether there is need for official notification [3]. WHO encourages countries to use its standard case definition for the reporting of cholera cases to ensure consistency, which states that a case of cholera should be suspected when (a) in an area where the disease is not known to be present, a patient aged five years or more develops severe dehydration or dies from AWD, and (b) in an area where there is a cholera epidemic, a patient aged five years or more develops AWD, with or without vomiting. A case of cholera is confirmed when *Vibrio cholerae* O1 or O139 is isolated by culture or PCR from any patient with diarrhoea [13]. WHO collates the cholera cases and deaths reported by the countries and publishes it in the Weekly Epidemiological Record (WER) every year.

However, in mild-to-moderate cholera infections, the symptoms may not be distinguishable from other causes of gastroenteritis. There is therefore considerable overlap between signs and symptoms of cholera with those of other diarrhoeal infections such as enterotoxigenic *Escherichia coli* (ETEC). As a result, real cholera cases might not even be suspected, or, alternatively, cases of AWD caused by other enteric pathogens be reported as cholera cases when laboratory confirmation is not available or feasible for all suspected cases (outbreak situations).

3. Problems with existing data sources/estimation systems

3.1. How close to reality is the calculated burden? Gaps in data

Cholera continues to be an under-recognized and under-reported problem in many endemic countries across the world. Several factors contribute to this underreporting, such as some countries reporting only laboratory-confirmed cases and other countries not reporting cases at all [14]. An example is the cholera-endemic state of Odisha, in India, which reported 740,000 cases and 235 deaths due to acute diarrhoeal diseases but no cases of cholera in the National Health Profile 2012 [15], whereas there were at least 77 cholera cases and five deaths in the state in 2012, according to numbers reported to IDSP [16]. A study identified several cholera cases and epidemics in the Indian Subcontinent and Southeast Asia through extensive literature searches which were not reported to WHO [14]. One reason is that authorities are reluctant to acknowledge and report cholera cases, which would reflect on the inadequacies in water and sanitation infrastructure in their areas [17]. Also, many cholera deaths occur before patients can reach treatment facilities and are therefore not captured in the hospital-based surveillance systems [4]. Finally, patients may not seek health advice when symptoms are mild [14]. Conversely, discrepancies and over-reporting might also arise due to reporting of clinically suspect cases, especially during outbreaks, when it is common practice to identify and report cholera cases on the basis of clinical signs and symptoms without laboratory confirmation [18]. Situations of humanitarian emergencies and internal conflicts might also result in over reporting due to destabilized diagnostic and confirmatory processes, such as speculated to have occurred in the ongoing cholera epidemic in Yemen, in which 1,103,683 suspected cholera cases were reported [19].

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3.2. Barriers in countries' laboratory capacities

In many cholera-prone settings, the ability to use culture-based diagnostic approaches is restricted by insufficient or lack of appropriate laboratory capacity. This constraint is determined by problems in staff management, lack of trained laboratory staff, unavailability of laboratory supplies, challenges with storage and transport of samples, unreliable reporting and lack of overall financial support. In particular, culture confirmation is rarely accessible in peripheral health care facilities where most cholera patients present for treatment and management. These challenges result in delays in outbreak detection and in implementation of control measures, which in turn leads to higher morbidity and mortality. This also limits the capacities for outbreak investigations and for the monitoring of *Vibrio cholerae* circulating strains, including capacity for the detection of the emergence of new strains that may require immediate changes in response (e.g. adapted antibiotics or vaccines).

3.3. Political and economic barriers to reporting

The fear of economic sanctions (ban on importation) that could lead to loss of trade, and travel restrictions that could impact tourism are major barriers to accurate reporting of cholera by countries. Although WHO guidelines regarding travel and trade practices advise against embargoes on imports of food from cholera-affected countries, restrictions on movements of people, requirement of proof of vaccination in case of international travel or chemoprophylaxis for travellers going to or from cholera-affected areas [20], many affected countries in the past have been subjected to exaggerated discriminatory practices by the international community in the wake of outbreak, as observed in the case of the 1991 cholera epidemic in Peru and 1994 plague outbreak in India [21]. Both countries were subjected to food trade embargoes, and often travellers from these nations were denied entry at their destinations or subjected to quarantine, against the WHO guidelines. This age-old dilemma needs to be addressed soon, perhaps by "rewarding" the countries for better reporting with technical support and improved access to vaccines.

For many endemic countries, being labelled as the origin of cholera transmission into non-endemic countries is another barrier to reporting. One study identified that most cases imported into the UK originated from the Indian subcontinent, with 32% of cases from 1990 to 2003 originating in India, 29% in Pakistan and 7% in Thailand. For 2004, the study identified 93 imported cases of cholera worldwide, and among the cases in which a travel destination could be determined, it was reported that India was the origin for 38 cases, the Philippines 36 cases and Thailand 11 cases [14]. These data cause concern and fear of bias in the countries identified as sources of spread of cholera and increase reluctance for reporting cholera.

3.4. Examples of gaps in the current estimation

A number of studies have attempted to identify the gaps in reported and estimated cholera burden data (Table 1).

Studies at subnational levels also present a similar picture. For example, a review reported that the number of cholera confirmed cases admitted at the Infectious Diseases Hospital in Kolkata, West Bengal, India were 3823, 7828 and 3242 for the cholera seasons 2007–2008, 2008–2009 and 2009–2010, respectively, which were higher than the figures reported by India for the entire country to WHO for the same period (2635, 2680 and 5155 for the years 2007, 2008 and 2010, respectively – no data reported in 2009) [15].

In case of endemic countries with large population, caution should be taken when extrapolating figures from small-scale studies. In one study which reported estimated global burden figures by supplementing WHO-reported figures from other sources, the authors note that China, India and Indonesia had to be treated differently from other countries in the estimate calculations because of their large populations, the risk of cholera being heterogeneous and their reporting structures organized by state or province. This is important because these countries greatly influence the global burden estimates and inclusion of entire populations (as done for other countries) could artificially increase the global estimates of at-risk population and cholera incidence [4].

4. Impact of gaps on burden calculation

4.1. Effect on economic analysis

The true impact of a disease has to be viewed in a broader framework, not just through the number of patients affected or number of deaths caused. Cholera is a disease of poverty; therefore, cholera incidence is a key indicator of social equity and development [29], and the economic consequences of cholera need to be analysed exhaustively to understand the full impact of the disease. For example, one study reported the impact of cholera on gross domestic product of Bangladesh, Mozambique, India and South Africa in the first year of a cholera outbreak, determined through economic modelling, to be greater than 2% [30]. However, because of unavailability of accurate burden estimates, a global economic analysis is difficult to undertake. A 2008 study calculated economic cost of cholera in Africa, based on 125,018 cases reported to WHO by the countries of the WHO Africa region. The authors reported a total economic loss of US$39–64.2 million for the region based on the reported incidence figures [31]. However, cholera estimates from another study in 2012 suggest cholera incidence in the region to be 1,411,453 for the same year [32]. These large differences between reported and estimated numbers highlight the extent to which economic analysis studies are hampered by lack of accurate burden data.

Table 1
Examples of under-reporting of cholera in selected countries in Asia.

Country	Time period	Reported number of cases	Estimated/suspected number of cases	Reference
India	1997–2006	37,783	222,038	[17]
Philippines	2008	0	2981	[22]
	2009	0	5645	[22]
	2010	33	5745	[22,23]
	2011	120	9769	[22,24]
	2012	1864	12,847	[22,25]
	2013	6	5084	[22,26]
Pakistan	2010–2014	3122	20,880	[23–28]

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4.2. Effect on deployment of interventions

National and regional-level decision-making by governments on planning strategies and interventions require compelling and detailed evidence of burden, including geographic distribution, incidence and severity of disease, in order to deploy targeted resources. However, transmission of cholera is extremely heterogeneous and is dependent on population density, socioeconomic differences and geographic and environmental features [33]. The heterogeneity of cholera and gaps in surveillance data make identifying target areas difficult, further contributing to the delay in deployment and reduced efficiency of interventions adapted for control.

4.3. Effect on demand generation for interventions

Demand for and uptake of effective and available interventions, such as the oral cholera vaccines (OCV), is dependent on availability of reliable burden data, lack of which impedes targeting of interventions, estimation of needs and appropriate communication to manufacturers for adjusting their production capacity. Studies in India exploring the reason for lack of demand for OCV and its adoption into national programme have repeatedly identified the reason that the disease is perceived to be non-existent or occurring in very small proportions within the country [15,34]. As a result, despite availability of a cost-effective, WHO-prequalified OCV that is manufactured in the country, it is not yet available through the country's public health programme, to reach populations where it is most needed. Studies that could analyse impact of OCV interventions and might provide additional evidence supporting their adoption are also impeded by the existing gaps in the surveillance systems and reporting.

5. Possible solutions for more accurate estimations

5.1. Strengthen existing surveillance systems

There is an urgent need to improve the epidemiological and laboratory surveillance for cholera at country level, to inform development and deployment of control and preventive measures and monitor their impact. A few initiatives have already been instigated to support cholera surveillance efforts in cholera-endemic countries, which include Africhol by Agence de Médecine Préventive, the Institut Pasteur International Network and the Enteric Diseases Laboratory Branch of the US Centers for Disease Prevention and Control (CDC).

The reinforcement of cholera surveillance in countries should target their capacity to diagnose and report cholera disease in a reliable and timely manner; enhance their investigation and response timeliness; improve the monitoring of *Vibrio cholerae* strains; and provide more accurate estimates of disease burden and geographical spread to best guide control interventions. This could be achieved by enhancing the capacity of national laboratories to extend the detection and diagnosis of cholera disease to the local level and to foster cross-border approaches by linking into regional networks and facilitating information sharing at the national and international level.

5.2. Supplement data from countries with other sources

Apart from hospital-based disease surveillance and reporting, other methods to augment burden data estimates need to be explored. In one study from within India, authors searched PubMed with key phrases to identify published literature reporting cholera within India, and supplemented it with reports published by Field Epidemiology Training Program of the National Institute of Epidemiology in Chennai, India, and annual reports of the National Institute of Cholera and Enteric Diseases (NICED) in Kolkata, India, which are institutes specializing in cholera and other diarrhoeal diseases in India [17]. Another study estimated global burden by sourcing cholera reports from literature searches of PubMed, as well as from reports of alternative disease monitoring systems, such as the Program for Monitoring Emerging Diseases (ProMED) and Global Infectious Disease and Epidemiology Network (GIDEON) [32]. An update to this study involved using a spatial regression model to predict an incidence rate of cholera and defining countries as endemic, non-endemic or cholera-free based on this [4]. Another study in Bangladesh attempted tracking cholera, especially with a view to early detection of outbreaks, by tracking sales of pharmacy-purchased oral rehydration solution (ORS) and found a significant correlation between ORS sales and hospitalized cholera [35].

Mathematical modelling is another method applied to retrospectively analyse and understand cholera epidemiology and its complex transmission dynamics as well as prospectively predict the evolution of cholera outbreaks and impact of interventions. It is used as a complementary tool to field epidemiology and statistical analysis [36]. However, cholera modelling faces the challenges of insufficient data for calibrating the models, the need to tailor models for different outbreak scenarios [33], uncertainties in model structure and parameter estimates, and spatial heterogeneity, which is intrinsic to cholera [37], and is therefore not used extensively so far in burden estimations.

5.3. Other alternative measures

Culture of the bacteria or its detection by PCR remains essential for the confirmation of the presence of the etiologic agent of cholera. At the same time, reliable cholera RDTs are critical to quickly alert medical authorities to the start of a potential outbreak. The value of cholera RDTs is particularly relevant for the resource-constrained settings where cholera is most likely to occur and access to formal laboratory services is limited or absent.

However, while there are several cholera RDTs on the market with others under development, their diagnostic accuracy has been irregular or there are concerns regarding product design. There is a need for developing new, better performing RDTs, which are usable at the periphery. WHO has undertaken development of a Target Product Profile for RDTs for cholera diagnosis, to help shape the development and production of affordable, high-quality RDTs, which can be validated and pre-qualified by WHO, allowing early detection of cholera outbreaks.

6. Conclusion

The year 2017 marked 200 years since the start of the first cholera pandemic when the disease was first noted by British troops in India. Cholera has been eliminated from Europe and Americas for decades, yet today the estimated cases and deaths due to cholera per year are in large numbers. Access to water and sanitation is a basic human right [38], yet 663 million people still lack access to improved drinking water sources, over 2 billion drink water from sources that are faecally contaminated and 2.4 billion are without improved sanitation facilities worldwide, and are exposed to the risk of waterborne diseases in general, including cholera [39]. The persistence of cholera reflects weaknesses of water and sanitation infrastructures and services, high-risk hygiene and social practices, shortcomings of surveillance and health care systems, lack of trained workforces to support outbreak response and limited access to OCVs.

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There is also growing evidence suggesting that climate change and urbanization are likely to play an increasing role in the global persistence of cholera. In Africa alone, the population is expected to double by 2050 and quadruple by 2100. If today 40 million people are at risk of cholera in Africa, the global burden of disease will increase as population grows. Rapid urbanization and population growth present increasing and competing demands for water and it is evident that the risk for cholera would increase without timely intervention. The foundation for any effective control or response measure for cholera largely depends on availability of reliable epidemiological data. It is imperative therefore that immediate steps are taken in strengthening surveillance and reporting systems globally, especially in cholera-prone and resource-limited areas, where it will enable countries to articulate their demand for resources more accurately.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

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